

Blockchain-assisted trusted computing and communication resource allocation strategy for industrial Internet of Things

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ABSTRACT

The Industrial Internet of Things (IIoT) has introduced digitalization and intelligence to the industrial field. However, the terminal in IIoT has a low computing power and limited storage resources, thus it is necessary to calculate and unload the data generated by the terminal and to allocate resources. Aiming to resolve the issues of computation offloading and resource allocation in IIoT, a blockchain-assisted data communication and computing service strategy of IIoT was constructed in this study. First, a blockchain-assisted computing and communication resource allocation model in IIoT was established, which embeds resource allocation and computation offloading into a block-chain to ensure the credibility of data during the process. In order to find the minimum value of the weighted sum of system energy cost, as well as the delay in blockchain-assisted resource allocation and computation offloading, a Levy flight-enhanced JAYA optimization algorithm (p-JAYA) based on blockchain resource allocation was proposed, which was embedded into the blockchain system to ensure the security of the data when allocating resources. Simultaneously, Levy flight was used to modify the algorithm to improve the convergence speed of the algorithm. The simulation results demonstrated that the Levy flight-enhanced JAYA optimization algorithm consumed 21% of the energy compared with the genetic algorithm, and 17.7% energy compared with the particle swarm optimization algorithm. The load balancing ability of the Levy flight-enhanced JAYA optimization algorithm is relative to the particle swarm optimization and genetic algorithms, which significantly increases the energy and delay consumption in the resource allocation of IIoT and improves the system performance.

1. Introduction

Considering the increasing popularity of smart devices and The swift advancement of 5G communication technology, the industrial Internet of Things (IIoT) has become a new paradigm and is widely used in many fields, gaining significant attention from academia and the industry [1–5]. There are numerous terminal devices in IIoT, many of which produce a significant volume of data, resulting in the rapid growth of data traffic in IIoT. However, the computing resources, storage resources and other resources of IIoT terminals are limited, making it impossible to efficiently complete the processing of the significant amount of data in a timely manner. Computation offloading and the resource allocation of data in IIoT significantly simplifies data processing by IIoT, reduces

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Table 1
Differences between this paper and previous studies.

Reference	D2D transmission between terminals	Blockchain	Communication interference	Sharing of surplus computing equipment resources	Algorithm
[9]	✓	×	×	✓	Learning-BasedCo-Offloading
[10]	×	×	×	✓	TCDA-Allocation
[11]	×	×	✓	✓	PSO
[13]	×	✓	×	✓	DRL
[23]	×	✓	×	×	SCA
Our	✓	✓	✓	✓	p-JAVA

the consumption of the terminal equipment resources [6–8], and improves the efficiency of data processing. Therefore, studying resource allocation in Industrial Internet of Things (IIoT) is of utmost importance.

Numerous scholars have conducted extensive research regarding resource allocation and computation offloading in IIoT. Dai et al. [9] proposed a co-offloading algorithm that combines D2D and edge computing for resource allocation in IIoT. The results demonstrated that the proposed algorithm can observe and analyze the system costs of potential edge nodes, then choose the most efficient one without relying on complete offloading information. Ma et al. [10] proposed a combined double-auction mechanism to ensure the authenticity and budget balance under local constraints in IoT resource allocation. This mechanism integrates the filling concept with an effective pricing strategy to ensure the performance under a constrained MEC. Lu et al. [11] proposed an evolutionary algorithm based on a Bayesian network (BNEA) to optimize the task allocation strategy considering the main problem of task resource allocation in IIoT. The results demonstrated that the response time of the BNEA can be delivered to the destination with minimal delay using the corresponding task allocation strategy. Deng et al. [12] applying the Q-learning algorithm for making discrete partial offloading decisions, aiming to minimize the delay in offloading services in the resource allocation of IIoT. The results demonstrated that the Q-learning scheme was delayed by 23%. Yang et al. [13] expressed the resource allocation problem in IIoT as a Markov decision, considering the minimization of energy consumption as the optimization goal and using deep reinforcement learning to solve the optimization problem. The results indicated that the entire system consumed a significant amount of energy. Gong et al. [14] addressed the resource allocation problem of IIoT, aiming to reduce the combined weight of delay and energy consumption using a new isobaric action generation technology and an adaptive action aggregation updating strategy to decompose the task offload; the algorithm demonstrated a quasi-optimal system performance. In the aforementioned studies, the credibility of data in IIoT devices during resource allocation and computation offloading is an issue.

Many scholars have conducted research to solve the problem of maintaining data credibility in the resource allocation of IIoT. Liu et al. [15] proposed a novel performance enhancement framework for an Industrial Internet of Things (IIoT) system based on blockchain and deep reinforcement learning (DRL). The purpose of the study was to develop a delay and security evaluation system. The results demonstrated that the framework can effectively improve the performance of an IIoT system that supports blockchain. Lu et al. Also using blockchain to solve the problem of industrial IoT resource scheduling is the literature [16–18], which establishes a blockchain-based resource allocation and sharing framework, and then uses deep learning to solve the related problems. [19] proposed a federated learning framework supported by blockchain running in a DTWN to address challenges such as unreliable communication channels, resource constraints, and distrust among users within the context of Industrial Internet of Things (IIoT). The results demonstrated that this framework had a higher efficiency and lower cost than benchmark learning. Lin et al. Similarly, in order to solve the problems of unreliable communication channels, limited resources and lack of trust among users in the industrial Internet of Things, literatures [20–22] also put forward their own solutions. [23] proposed a UAV-assisted IIoT system based on a private blockchain to solve the problem of IIoT devices usually consuming excessive energy to generate a blockchain. Compared to several other benchmark schemes, the simulation results demonstrated that this scheme can significantly reduce the energy consumption required for generating blockchains in IIoT. In addition, references are also made [24–26]. The aforementioned research only introduces blockchain to ensure the credibility of data in the resource allocation of IIoT, but lacks the optimization of the system loss introduced by the blockchain itself. Furthermore, it does not distinguish between the tasks generated by different devices in IIoT, which leads to a low system efficiency. The importance of tasks with a relatively high energy consumption and delays was not reflected. In order to further clarify the gap between this paper and previous studies, we compare between terminals in terms of D2D transmission, blockchain, communication interference, and residual resource sharing of computing devices, as shown in Table 1:

To solve the aforementioned problems, in this study, a blockchain-assisted computing and communication resource allocation model that uses blockchain security to ensure the authenticity of data in resource allocation was first constructed for IIoT. Second, a model of blockchain-aided computing and communication resource allocation for IIoT was derived, which combines the blockchain loss with the system loss of the resource allocation and minimizes the energy used and delay. To comprehensively solve the optimization problem of loss caused by the blockchain and that by the resource allocation, a p-JAYA optimization algorithm based on blockchain resource allocation was proposed, which decomposes tasks using a Bayesian network structure and determines their importance. To avoid falling into the local optimum solution and balance resource allocation, a probability-based updating strategy and a non-dominated sorting load-balancing strategy were introduced. Finally, the simulation results demonstrated that the p-JAYA optimization algorithm based on a Bayesian network has a stronger optimization ability, as well as a higher data credibility and load-balancing ability in resource allocation.

This study primarily contributes in the following ways:

Table 2
Glossary of terms used in the text.

Symbol	Meaning
TTE	Industrial IoT terminal set
TR	Data tasks generated by the terminal
STR	Size of the data task
ECS	Set of edge servers
RRC	Resource capacity of edge servers
VTs	Data transfer rate from terminal to master server
TTS	Data transfer time from terminal to master server
DPT	Main server data processing time
ECDP	Energy consumption of the main server for processing data
TCTE	Total energy consumption of the main server for processing data
TPPN	Transmission time for master server to transfer data to neighboring servers
DPTN	Data processing time of neighboring servers
ECN	Data processing energy consumption of neighboring servers
TCN	Total data processing loss on neighboring servers
TTPT	Processing time for terminal cluster processing data
TPTE	Energy consumption of terminal clusters for processing data
TPTC	Total loss of data processed by the terminal cluster
TPPC	Data transfer time from the main server to the cloud server
DPTC	Data processing time for cloud servers
ECC	Energy consumption of cloud servers for processing data
TCC	Total loss of data processed by the cloud server
TSC	Total loss of system processing tasks

- (1) A blockchain system was introduced into the computing and communication resource allocation of IIoT, which ensures the credibility of data and integrates the loss problem caused by the blockchain system into the overall optimization scheme, thus improving the overall system loss and efficiency.
- (2) Considering the different attributes of tasks generated by various IIoT devices in practice, a Bayesian network was used to decompose tasks and obtain task sequences based on their importance. This method not only reduced the overall system energy consumption and latency but also maximized the needs of different tasks.
- (3) A Bayesian-based p-JAYA optimization algorithm was proposed. The algorithm adopts a strategy utilizing probabilities to prevent getting stuck in a local optimum, and a non-dominated sorting load-balancing strategy to balance resource allocation

For ease of reading this article, common terms used in this article are shown in [Table 2](#).

2. System model

A model diagram of the computing and communication resource-allocation algorithm based on blockchain in IIoT is shown in [Fig. 1](#), which consists of three layers. The first layer is the IIoT device layer, which mainly includes the IIoT devices, which are represented as follows:

$TTE = \{TE_1, TE_2, TE_3, \dots, TE_j, \dots, TE_{m \times n}\}$, where m and n indicate the number of edge servers and IIoT devices covered by each server, respectively. The data tasks generated by the terminals in an IoT system are expressed as follows: $TR_{ij} = \{STR_{ij}, CRT_{ij}, TM_{ij}, TI_{ij}, IM_{ij}, F_{ij}\}$. where STR_{ij} represents the size of the data, CRT_{ij} is the computational resource needed to process the data, TM_{ij} is the data TR_{ij} and the arrival time F_{ij} indicates whether the task can be split. In IIoT, the data generated by the IoT terminal must be completed in a short time. TI_{ij} is the maximum delay allowed for completing the transaction; the importance of a transaction is denoted by IM_{ij} . The second layer is an edge-computing server layer, which is represented as follows: $ECS = \{BS_1, BS_2, \dots, BS_i, \dots, BS_m\}$. The resource capacity of the K -th edge server is expressed as follows:

$RRC_k = \{RSC_k, RCR_k, RCOR_k\}$, where $RRC_k = \{RSC_k, RCR_k, RCOR_k\}$ is the storage resource, RCR_k is the computing resource, and $RCOR_k$ is the communication resource of the k th edge computing server. The third layer is the cloud server layer. The storage, communication, and computing resources of the cloud server are infinite and are mainly used to manage tasks with significant computations but low delay requirements.

Data transmission between devices covered by the edge computing server BS_i is distributed by a D2D communication channel; the data and signaling between devices and base stations interact via the 5G network. Terminal and base stations covered by the same server interact by orthogonal frequency-division multiplexing technology without interference between one another. Adjacent servers can use the same spectrum resources; therefore, interference may occur between adjacent servers.

The data information generated by the terminal TE_j^i is first uploaded to an edge computing server (master server) covering the terminal; the edge computing server then executes a task allocation algorithm to allocate resources for the data generated by the terminal to complete the data processing. There are four possible methods for data processing, which are conducted by using the following: (1) main server, (2) neighboring server, (3) terminal cluster, and (4) uploading data to the cloud for processing. This study uses the following variables:

$$tkpro = \left\{ pm_{ij}^k \mid pm_{ij}^k \in \left\{ pm_{ij}^{ec}, pm_{ij}^{nc}, pm_{ij}^{dc}, pm_{ij}^{rc} \right\}, \right. \\ \left. pm_{ij}^k \in \{0, 1\}, i \in N, j \in M, k \in \mathbf{K} = \{ec, nc, dc, rc\} \right\} \quad (1)$$

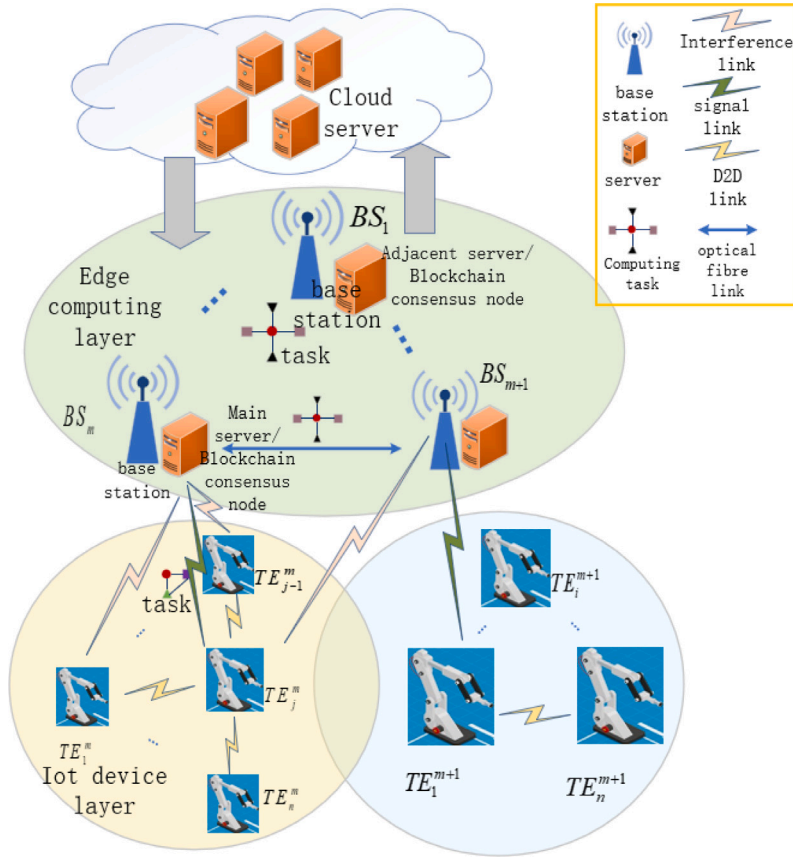


Fig. 1. Computing and communication resource-allocation algorithm system model based on a blockchain system in industrial Internet of Things (IIoT).

represents task processing, where $pm_{ij}^k = 1$ and $pm_{ij}^k = 0$ indicate that the data was executed and not executed under the k th execution, respectively. For each type of execution, a utility function was established in this study to comprehensively measure the value and cost.

- (1) Primary server execution: The cost of using the master server to process data includes the time required to upload data from the terminal to the master server. The frequency band resource provided for the data output between the terminal and main server is set as BTS_{ij} , and the transmission rate at which the data is uploaded from the terminal to the master server can be expressed as follows:

$$VTS_{ij} = BTS_{ij} \log_2 (1 + (PW \cdot TG)/\sigma^2) \quad (2)$$

Here, PW is the terminal transmission power, TG is the transmission gain, and σ^2 is the white Gaussian noise in the channel. The time required for data transmission from the terminal to the main server is referred to as follows:

$$TTS_{ij} = \frac{STR_{ij}}{VTS} = \frac{STR_{ij}}{BTS_{ij} \log_2 (1 + (PW \cdot TG)/\sigma^2)} \quad (3)$$

Assuming that the CPU frequency of the master server is MSF_j , the data processing speed of the unit cycle is DPR_j , and the data processing time is

$$DPT_{ij} = \frac{STR_{ij}}{MSF_j \cdot DPR_j} \quad (4)$$

Assuming that the CPU of the main server consumes an energy per cycle $ECPC_j$ and the energy consumption coefficient is β_{ms} , the energy consumed by data processing is as follows:

$$ECDP_j = \beta_{ms} \cdot STR_{ij} \cdot (MSF_j)^2 \cdot ECPC_j \quad (5)$$

Generally, the cost of data processing by the master server is

$$\begin{aligned} TCTE_j = & (1 - \xi_{ms}) (ECDP_j + PW \cdot TTS_{ij}) \\ & + \xi_{ms} (TTS_{ij} + DPT_{ij}) \end{aligned} \quad (6)$$

where ξ_{ms} represents the weight coefficient.

- (2) Adjacent server execution: The data path is from the terminal to the primary server, and from the primary server to the neighboring server. The transfer time from the primary server to the neighboring server is as follows:

$$\begin{aligned} TTPN_{ij} &= \frac{STR_{ij}}{VPN_{ij}} \\ &= \frac{STR_{ij}}{BPN_{ij} \log_2 (1 + (PWP \cdot TGPN)/\sigma^2)} \end{aligned} \quad (7)$$

Here,

$$VPN_{ij} = BPN_{ij} \log_2 (1 + (PWP \cdot TGPN)/\sigma^2) \quad (8)$$

denotes the data transmission rate from the master server to the adjacent server, BPN_{ij} denotes the data transmission bandwidth between the master server and adjacent server, PWP denotes the transmission power of the master server, and $TGPN$ denotes the channel gain between the master server and adjacent server. The data processing time of the adjacent servers is as follows:

$$DPTN_{jk} = \frac{STR_{ijk}}{MSFN_k \cdot DPRN_k} \quad (9)$$

The energy consumed by executing data on the neighboring servers is as follows:

$$ECN_j = \beta_k \cdot STR_{ijk} \cdot MSFN_k^2 \cdot ECCN_k \quad (10)$$

where $MSFN_k$ is the CPU frequency of the adjacent server, $DPRN_k$ is the data processing speed of the adjacent server, β_k is the energy consumption coefficient of the adjacent server, and $ECCN_k$ is the energy consumption per cycle of the adjacent server. The total cost of the processing data on the proximity is as follows:

$$\begin{aligned} TCN_k &= (1 - \xi_k) (PW \cdot TTS_{ij} + TTPN_{ij} \cdot PWP + ECN_j) \\ &\quad + \xi_k (TTS_{ij} + TTPN_{ij} + DPTN_{jk}) \end{aligned} \quad (11)$$

where ξ_k represents the weight coefficient of the time and energy.

- (3) Terminal equipment cluster execution: Owing to the low storage and computing capacity of the terminal, the task must be decomposed, and each small task is executed by one terminal. First, it is determined whether the flag bit F_{ij} of the data is 1, which indicates that the data can be split; if it is not 1, the data cannot be split. If the data can be split, then we assume that the task is split into $\{sub_{ij}^1, sub_{ij}^2, sub_{ij}^3, \dots, sub_{ij}^\delta\}$, where δ represents the number of subtasks. Subtask sub_{ij}^k is executed by the terminal device TE_j^n , assuming that the size of the subtask is $subs_{ij}^k$. The CPU frequency of the n th terminal is TCF_j^n . The data processing speed of the n th terminal is TPS_j^n . When $n = i$, the transmission time is 0, and the processing time is:

$$TPT_n^k = \frac{subs_{ij}^k}{TCF_j^n \cdot TPS_j^n} \quad (12)$$

The energy consumed in processing the data is given by

$$TPDN_n^k = \beta_n \cdot subs_{ij}^k \cdot (TCF_j^n)^2 \cdot TCCN_n \quad (13)$$

where β_n is the energy consumption coefficient of the terminal TE_j^n , and $TCCN_n$ is the energy consumption per cycle of the terminal TE_j^n . When $n \neq i$, the data transmission time is given by

$$\begin{aligned} TDDTT_{i \rightarrow n}^k &= \frac{subs_{ij}^k}{VTT_{i \rightarrow n}^k} \\ &= \frac{subs_{ij}^k}{BTT_{i \rightarrow n}^k \cdot \log_2 (1 + (PWTT \cdot TGTT)/\sigma^2)} \end{aligned} \quad (14)$$

Here, $VTT_{i \rightarrow n}^k$ is the data transmission rate between terminals, $BTT_{i \rightarrow n}^k$ is the data transmission bandwidth between terminals, $PWTT$ is the terminal transmission power, $TGTT$ is the channel gain between terminals, and σ^2 is the Gaussian white noise channel. The total time spent on executing the terminal device cluster is given by

$$TTPPT_j^i = \max \{TDDTT_{i \rightarrow n}^k + TPT_n^k\} \quad (15)$$

The total energy consumed is

$$TPTE_i = \sum_{k=1}^{\delta} TPDN_n^k \quad (16)$$

The total cost of processing data by the terminal is

$$\begin{aligned} TPTC_j &= \xi_{dc} TTPPT_j^i + (1 - \xi_{dc}) TPTE_i \\ &= \xi_{dc} \max \{TDDTT_{i \rightarrow n}^k + TPT_n^k\} + (1 - \xi_{dc}) \sum_{j=1}^{\delta} TPT_n^k \end{aligned} \quad (17)$$

where ξ_{dc} is the weight coefficient.

- (4) Cloud server execution: The data transmission path is from the terminal to the main server, and from the main server to the cloud server. The transmission time of the data from the main server to the cloud server is denoted by

$$TTPC_{ij} = \frac{STR_{ij}}{VPC_{ij}} = \frac{STR_{ij}}{BPC_{ij} \log_2(1 + PWPC \cdot TGPC) / \sigma_c^2} \quad (18)$$

Here,

$$VPC_{ij} = BPC_{ij} \log_2(1 + PWPC \cdot TGPC) / \sigma_c^2 \quad (19)$$

is the data transfer rate of the data transferred from the main server to the cloud server, BPC_{ij} is the data transmission bandwidth from the main server to the cloud server, $PWPC$ is the transmission power of the main server, $TGPC$ is the channel gain of the data transmitted from the main server to the cloud server, and σ_c^2 is the Gaussian white noise. The data processing time of the cloud server is given by

$$DPTC_{jc} = \frac{STR_{jc}}{MSFC_c \cdot DPRC_c} \quad (20)$$

The energy consumed when data is executed on the cloud server is

$$ECC_j = \beta_c \cdot STR_j \cdot MSFC_c^2 \cdot ECCC_c \quad (21)$$

where $MSFC_c$ is the CPU frequency of the cloud server, $DPRC_c$ is the data processing speed, β_c is the energy consumption coefficient of the cloud server, and $ECCC_c$ is the energy consumption per cycle of the cloud server. The total cost of data processing on a cloud server is as follows:

$$TCC_c = (1 - \xi_c) (PW \cdot TTS_{ij} + TTPC_{ij} \cdot PWPC + ECC_j) + \xi_c (TTS_{ij} + TTPC_{ij} + DPTC_{jc}) \quad (22)$$

where ξ_c represents the weight coefficient between the time and energy.

Based on the aforementioned analysis, the cost consumed by the system when processing task TR_{ij} is as follows:

$$TSC_{ij} = \gamma_{ms}^{ij} TCTE_j + \gamma_k^{ij} TCN_k + \gamma_{dc}^{ij} TPTC_j + \gamma_c^{ij} TCC_c \quad (23)$$

Here, $\gamma_{ms}^{ij}, \gamma_k^{ij}, \gamma_{dc}^{ij}, \gamma_c^{ij} \in \{0, 1\}$, $\gamma_{ms}^{ij} + \gamma_k^{ij} + \gamma_{dc}^{ij} + \gamma_c^{ij} = 1$ indicates the weight factors of the main server, neighboring servers, device clusters, and cloud servers in the system consumption when executing task TR_{ij} respectively. The total system cost is given by

$$TSC = \sum_{i \in m} \sum_{j \in n} TSC_{ij} \quad (24)$$

The data processing methods that can be used for the computing tasks generated by an IoT terminal are related to the type of task. For example, delay sensitive tasks have significant time requirements and cannot be processed in the cloud. When the main server receives a task, it first determines the type of task and processing method that can be used. To illustrate this point, this study abstracted the tasks generated by IoT terminals. As previously indicated, for each task generated by the IoT terminal, we used

$$TR_{ij} = \{STR_{ij}, CRT_{ij}, TM_{ij}, TI_{ij}, IM_{ij}, F_{ij}\} \quad (25)$$

to represent each task; each parameter has been defined above. The following focuses on the IM_{ij} parameter of task TR_{ij} . The IM_{ij} parameter indicates the importance of the task. The tasks are divided into 4 types based on their importance, each of which can be handled in different ways. A low-level task can take more execution methods. The importance of a task is calculated as follows:

- (1) First, each task has an initial importance, denoted by SI_i , which is assigned to the task by each terminal when generating data. Its size depends on the degree of contribution of the task to the system or terminal, and whether the task is executed.
- (2) The dynamic importance of each task is calculated by the following formula:

$$DI_i = \eta_i (TI_{ij} - TP_{ij}) \quad (26)$$

where η_i is the remaining delay of the task and TP_{ij} is the time that has elapsed since the task was generated. The overall importance of the task is then expressed as follows:

$$GI_{ij} = \mu_{ts} SI_{ij} + \mu_{td} DI_{ij} \quad (27)$$

This study introduces blockchain, which has a sharing function, and uses its sharing ability to provide the relevant remaining resource information for the resource allocation process. It mainly includes the computing resources of the main server, adjacent servers, and terminal device cluster.

The design goal of the entire system is to achieve energy-saving and a real-time operation by completing the data-processing task. In this study, the weighted sum of the delay and energy are considered for achieving optimization, and the allocation of computing resources is used as the optimization variable. To better allocate computing resources, variables x_r , Y and Z were introduced, which were used as the decision variables of the optimization problem. Among these, $x_r \in \{r_1, r_2, r_3, r_4\}$ represents the task execution level.

For example, if $x_r = r_1$, the task execution level is r_1 , which is the highest level among the task execution levels. The task is of the greatest importance, thus it can only be executed by the main server. Level r_2 is a more important task execution mode, which can be executed by the main and adjacent servers. By analogy, r_3 can be executed by the main server, adjacent servers, and terminals. Lastly, r_4 indicates that the task can be executed by all the methods.

$$Y = (pm_{ij}^{ec}, pm_{ij}^{nc}, pm_{ij}^{dc}, pm_{ij}^{rc}) \quad (28)$$

indicates that the specific execution mode is selected from the execution mode field. For example, $pm_{ij}^{ec} = 1$ indicates that a task is executed on the main server, and the value of $pm_{ij}^{nc}, pm_{ij}^{dc}, pm_{ij}^{rc}$ in the other execution methods is zero, indicating that this execution method is not used. Z involves determining a specific executor in the execution mode selected by

$$Y = (pm_{ij}^{ec}, pm_{ij}^{nc}, pm_{ij}^{dc}, pm_{ij}^{rc}) \quad (29)$$

Namely, Z determines the device that will handle the task. The aforementioned optimization variables represent multilevel optimizations. To simplify the optimization problem, the optimization variables are redefined as follows:

$$\Phi = \{ \theta_{ij}^k \mid i \in N, j \in M, \theta_{ij}^k \in \{0, 1\} \} \quad (30)$$

where $\theta_{ij}^k = 1$ indicates that the task TR_{ij} is executed in the k th manner; otherwise, it is not executed in this manner. Based on the aforementioned analysis, the optimization problem established in this study is as follows: eq31 – 38

$$\min_{\{\theta_{ij}^k\}} TSC \quad (31)$$

$$\text{s.t. } \gamma_{ms}^{ij}, \gamma_k^{ij}, \gamma_{dc}^{ij}, \gamma_c^{ij} \in \{0, 1\}, \gamma_{ms}^{ij} + \gamma_k^{ij} + \gamma_{dc}^{ij} + \gamma_c^{ij} = 1 \quad (32)$$

$$\begin{aligned} pm_{ij}^{ec}, pm_{ij}^{nc}, pm_{ij}^{dc}, pm_{ij}^{rc} &\in \{0, 1\}, \\ pm_{ij}^{ec} + pm_{ij}^{nc} + pm_{ij}^{dc} + pm_{ij}^{rc} &= 1 \end{aligned} \quad (33)$$

$$\sum_{i \in M} \sum_{j \in N} CRT_{ij} \leq CRT \quad (34)$$

$$TTS_{ij} + DPT_{ij} \leq TI_{ij} \quad (35)$$

$$TTS_{ij} + TTPN_{ij} + DPTN_{jk} \leq TI_{ij} \quad (36)$$

$$TTPPT_n^k \leq TI_{ij} \quad (37)$$

$$TTS_{ij} + TTPC_{ij} + DPTC_{ij} \leq TI_{ij} \quad (38)$$

Eq. (31) is the optimization objective function, and Eq. (32) is the weight coefficient of the energy consumed by each execution mode when TR_{ij} is executed. Eq. (33) represents a mutually exclusive relationship between the execution modes, which indicates that each task can have only one execution mode. Eq. (34) indicates that the computing resources required for the execution of all the tasks cannot exceed those of the entire system. Eq. (35) indicates that if a task is executed on the main server, the execution time cannot exceed the maximum delay allowed by the task. Equations (36), (37), and (38) indicate that if a task is executed by a nearby server, terminal, or cloud server, respectively, the total task processing time cannot exceed the maximum delay allowed by the task.

Introducing a blockchain into an IoT system can increase its security. In this system, we must sign the terminal data, verify it on the blockchain, and encrypt the terminal request information. Therefore, this study uses the decentralized characteristics of the blockchain to establish a blockchain-based security solution as follows.

In a blockchain system, each terminal requires a globally unique identification, and the development of asymmetric cryptography provides a remedy for this issue. The public and private keys of asymmetric cryptography can uniquely identify a terminal; therefore, a public and private key pair for each terminal was established in this study, which is expressed as follows:

$$\{ PK_{d_i}, SK_{d_i}, Cert_{d_i} \} \quad (39)$$

where PK_{d_i} represents the public key of the i th terminal, and is used as the terminal address in the blockchain system to identify the source of the data. Furthermore, SK_{d_i} represents the private key of the i th terminal, which is generally used for signatures in the blockchain system. SK_{d_i} indicates the terminal from which the data are received. Signature verification using the corresponding public key of the terminal can verify the authenticity of the data. $Cert_{d_i}$ represents the certificate of the terminal, which can uniquely identify a terminal.

During the system initialization phase, each industrial IoT terminal uses public keys, certificates, and other information to generate account information. Account information is stored in the account information pool of the blockchain system such that the blockchain system can identify any data offloaded by the terminal. Upon request, information such as the fuel fees can be issued to terminals participating in the calculations.

For security, the computational offloading request information of the terminal must be encrypted and signed. The terminal request information is expressed as follows:

$$\text{req}_j^i = E \left(TR_{ij} \mid dh_{ij} \mid PK_{d_i} \mid \text{sig}_{d_i} \mid \text{Cert}_{d_i} \right) \quad (40)$$

Here,

$$\text{sig}_{d_i} = \text{sign}_{SK_{d_i}} (TR_{ij} \mid dh_{ij}) \quad (41)$$

indicates that the terminal uses its own private key SK_{d_i} to sign the data information TR_{ij} and data hash dh_{ij} .

In blockchain-assisted computing and communication resource allocation systems for IIoT, various resource carriers regularly call resource contracts to synchronize their remaining resources with the blockchain system. The remaining resource information is expressed as follows:

$$\text{msg}_i = E \left(RRX_i \mid PK_i \mid \text{sig}_i \mid \text{Cert}_i \mid tp \right) \quad (42)$$

where

$$RRX_i = \{RRD_i, RRE_i, RRC\} \quad (43)$$

represents the remaining resources of the i th terminal, i th edge server, or cloud server, and PK_i represents the public key identification of the i th terminal, i th edge server, or cloud server.

$$\text{sig}_i = \text{sign}_{SK_i} (RRX_i) \quad (44)$$

represents the signature information of the i th terminal, i th edge server, or cloud server using its own private key SK_i for the remaining resource RRX_i , Cert_i represents the certificate of the i th terminal, i th edge server, or cloud server. Furthermore, tp represents the timestamp at which the remaining resources are synchronized to the blockchain network, and the distributed storage and synchronization characteristics of the blockchain are used to quickly obtain the remaining global resource information during resource allocation, which is conducive to the optimization and solution of the optimization algorithm. After obtaining all the requested information indicated by

$$\text{req} = \left\{ \text{req}_j^i \mid i \in N, j \in M \right\} \quad (45)$$

and the remaining resource information indicated by

$$\text{msg} = \{ \text{msg}_i \mid i = d_i + e_i + c \} \quad (46)$$

in the entire system, this information can be submitted to the optimization algorithm to calculate the optimal offloading decision and resource allocation solution. The calculation offloading and resource allocation processes are then continued. After the assigned device $eq = \{eq_p \mid p \in \{d_i, e_i, c\}\}$ completes its offloading task, a transaction is generated to record the offloading event. A transaction is represented as follows:

$$\begin{aligned} \text{Tr}_{d_j^i \rightarrow eq_p} = & E \left(TR_{ij} \mid dh_{ij} \mid RS_{d_j^i \rightarrow eq_p} \mid \left(\text{sig}_{d_j^i}, \text{sig}_{eq_p} \right) \mid \right. \\ & \left. \left(PK_{d_j^i}, PK_{eq_p} \right) \mid \left(\text{addr}_{d_j^i}, \text{addr}_{eq_p} \right) \mid \text{gas} \mid tp \right) \end{aligned} \quad (47)$$

where d_i represents the i th device in the device set, e_i represents the i th edge server in the edge-computing server set, c represents the cloud server, and eq_p represents the device p , which can be a terminal, edge-computing server, or cloud server.

$$RS_{d_j^i \rightarrow eq_p} = \left(CTR_{d_j^i \rightarrow eq_p}, CPR_{d_j^i \rightarrow eq_p}, SGR_{d_j^i \rightarrow eq_p} \right) \quad (48)$$

represents the set of communication computing storage resources required to offload the task TR_{ij} generated at the terminal TE_j^i to device eq_p . $CTR_{d_j^i \rightarrow eq_p}$ represents the communication resources required when task TR_{ij} generated by terminal TE_j^i is offloaded to device eq_p , $CPR_{d_j^i \rightarrow eq_p}$ represents the computing resources required when task TR_{ij} generated by terminal TE_j^i is offloaded to device eq_p , and $SGR_{d_j^i \rightarrow eq_p}$ represents the storage resources required when task TR_{ij} generated by terminal TE_j^i is offloaded to device eq_p .

$$\text{sig}_{d_j^i} = \text{sign}_{SK_{d_j^i}} (TR_{ij} \mid dh_{ij} \mid RS_{d_j^i \rightarrow eq_p}) \quad (49)$$

indicates that terminal TE_j^i uses its own private key $SK_{d_j^i}$ to sign the task TR_{ij} generated by it, the data hash dh_{ij} , and resource set $RS_{d_j^i \rightarrow eq_p}$ used for uninstallation. The signature function is $\text{sign}_{SK_{d_j^i}}$.

$$\text{sig}_{eq_p} = \text{sign}_{SK_{eq_p}} (TR_{ij} \mid dh_{ij} \mid RS_{d_j^i \rightarrow eq_p}) \quad (50)$$

indicates that device eq_p uses its own private key SK_{eq_p} to sign the task TR_{ij} generated by terminal TE_j^i , the data hash dh_{ij} , and resource set $RS_{d_j^i \rightarrow eq_p}$ used for offload. The signature function is $\text{sign}_{SK_{eq_p}}$. $PK_{d_j^i}$ represents the public key of terminal TE_j^i , PK_{eq_p} represents the public key of device eq_p , and $\text{addr}_{d_j^i}$ represents the blockchain address of terminal TE_j^i , which is used for the

transfer. $addr_{eq_p}$ represents the blockchain address of device eq_p , gas represents the remuneration indicating that terminal TE_j^i must pay device eq_p for processing task TR_{ij} , and tp represents the transaction timestamp.

After the blockchain system was introduced, because additional requests and transaction data also need to be processed, which requires memory, CPU, and other computing resources, it is necessary to consider combining these data for optimization. Assuming that the data size of the request information req_j^i is lr_j^i , that of the remaining resource information msg_i is lm_j^i , and of the transaction $Tr_{d_j^i \rightarrow eq_p}^i$ is lt_j^i , then the following applies:

- (1) When the main server is executed, the time required to transmit data from the terminal to the main server is as follows:

$$TTS_{ij} = \frac{STR_{ij} + lr_j^i}{VTS} = \frac{STR_{ij} + lr_j^i}{BTS_{ij} \log_2(1 + (PW \cdot TG)/\sigma^2)} \quad (51)$$

The data processing time is as follows:

$$DPT_{ij} = \frac{STR_{ij} + lr_j^i + lt_j^i}{MSF_j \cdot DPR_j} \quad (52)$$

The energy consumed in data processing is

$$ECDP_j = \beta_{ms} \cdot (STR_{ij} + lr_j^i + lt_j^i) \cdot MSF_j^2 \cdot ECPC_j \quad (53)$$

- (2) When the neighboring server is executed, the transfer time from the primary server to the neighboring server is

$$\begin{aligned} TTPN_{ij} &= \frac{STR_{ij}}{VPN_{ij}} \\ &= \frac{STR_{ij}}{BPN_{ij} \log_2(1 + (PWP \cdot TGPN)/\sigma^2)} \end{aligned} \quad (54)$$

The data processing time of the proximity server is:

$$DPTN_{jk} = \frac{STR_{ijk} + lr_j^i}{MSFN_k \cdot DPRN_k} \quad (55)$$

The energy consumed by data execution on the nearby servers is:

$$\begin{aligned} ECN_j &= \beta_k \cdot (STR_{ijk} + lr_j^i) \cdot MSFN_k^2 \cdot ECCN_k \\ &+ \beta_{ms} \cdot lr_j^i \cdot MSF_j^2 \cdot ECPC_j \end{aligned} \quad (56)$$

- (3) When the terminal device executes, the processing time is

$$TPT_n^k = \frac{subs_{ij}^k + lt_j^i}{TCF_j^n \cdot TPS_j^n} \quad (57)$$

The energy consumed in the processing data is

$$TPDN_n^k = \beta_n \cdot (subs_{ij}^k + lt_j^i) \cdot (TCF_j^n)^2 \cdot TCCN_n \quad (58)$$

The total energy consumed is

$$TPTE_i = \sum_{k=1}^{\delta} TPDN_n^k + \beta_{ms} \cdot lr_j^i \cdot MSF_j^2 \cdot ECPC_j \quad (59)$$

- (4) When the cloud server executes, the data processing time of the cloud server is

$$DPTC_{jc} = \frac{STR_{jc} + lr_j^i}{MSFC_c \cdot DPRC_c} \quad (60)$$

The energy consumed when the data is executed on the cloud server is

$$\begin{aligned} ECC_j &= \beta_c \cdot (STR_j + lr_j^i) \cdot MSFC_c^2 \cdot ECCC_c \\ &+ \beta_m \cdot lr_j^i \cdot MSF_j^2 \cdot ECPC_j \end{aligned} \quad (61)$$

In the derivation of the formula, the other formulas remain unchanged, and the optimization objective function is rewritten as follows:

$$\min_{\{\theta_{ij}^*\}} TSC \quad (62)$$

$$\text{s.t. } \gamma_{ms}^{ij}, \gamma_k^{ij}, \gamma_{dc}^{ij}, \gamma_c^{ij} \in \{0, 1\}, \gamma_{ms}^{ij} + \gamma_k^{ij} + \gamma_{dc}^{ij} + \gamma_c^{ij} = 1 \quad (63)$$

$$\sum_{i \in M} \sum_{j \in N} CRT_{ij} \leq CRT \quad (64)$$

$$TTS_{ij} + DPT_{ij} \leq TI_{ij} \quad (65)$$

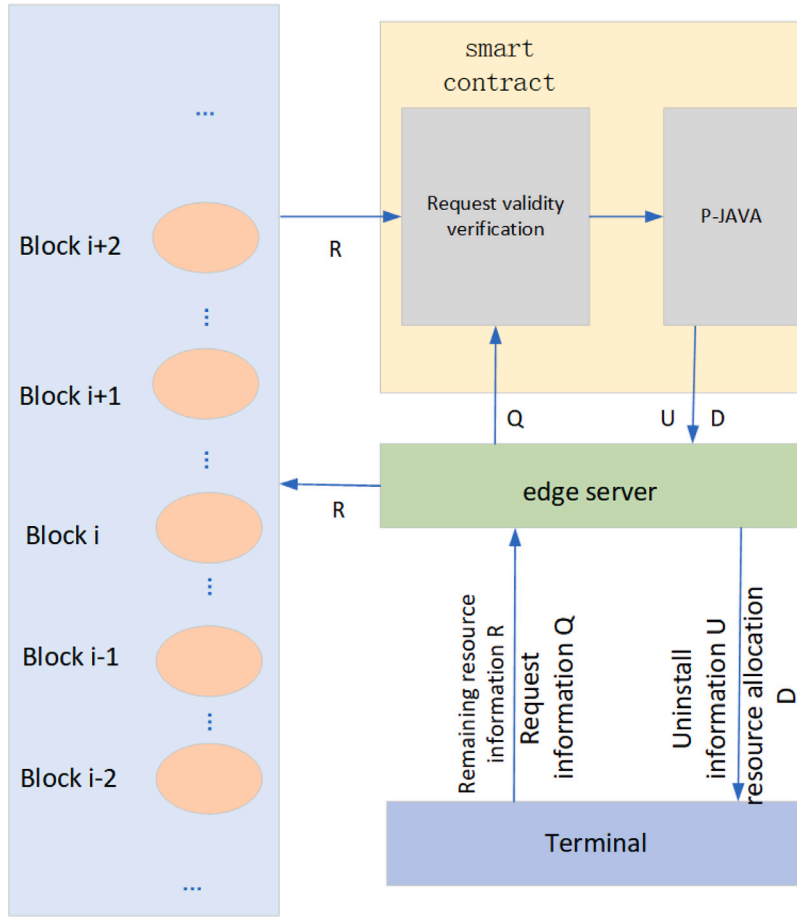


Fig. 2. Terminal task data processing block diagram.

$$TTS_{ij} + TTPN_{ij} + DPTN_{jk} \leq TI_{ij} \quad (66)$$

$$TTPN_{ij}^k \leq TI_{ij} \quad (67)$$

$$TTS_{ij} + TTPC_{ij} + DPTC_{ij} \leq TI_{ij} \quad (68)$$

$$\sum_{i \in M} \sum_{j \in N} TR_{ij} \leq CTR \quad (69)$$

Eq. (69) represents the storage-resource limit. Based on the aforementioned derivation, a blockchain-based IoT resource-allocation framework can be obtained, as shown in the figure below.

3. Levy flight-enhanced p-JAYA optimization algorithm based on blockchain resource allocation

The p-JAYA optimization algorithm is an efficient swarm intelligence optimization algorithm that can solve both constrained and unconstrained problems. Compared with other optimization algorithms, it has the advantage of having no algorithm control parameters, which significantly improves the optimization efficiency. The p-JAYA optimization algorithm is divided into four parts: initializing the population, updating individuals, greedy selection, and outputting an optimal solution.

(1) Initializing the population: The p-JAYA algorithm randomly initializes the population in the search space as follows:

$$\begin{aligned} Y_{i,j} &= Y_{i,j}^l + \text{rand}(0, 1) \times (Y_{i,j}^u - Y_{i,j}^l), \\ &= 1, 2, \dots, np, j = 1, 2, \dots, \text{dim} \end{aligned} \quad (70)$$

where np is the population size, dim is the number of variables, $Y_{i,j}^u$ and $Y_{i,j}^l$ are the upper and lower bounds of the search space, and $rand(0, 1)$ is a random number with a value range of $[0, 1]$. This initialization method is simple and easy to implement; however, its initial conditions are unstable, which can easily cause instability of the solution. If the initial conditions are limited to a certain area, the obtained solution can easily become a local optimal solution. To increase the diversity of the population and the ergodicity of the algorithm, a low-difference sequence was used in this study to initialize the p-JAYA algorithm. The Hammersley sequence is a widely used low-difference sequence, which presents a good uniformity of the initial conditions in a high-dimensional space.

- (2) Solution iteration: Each vector in the p-JAYA algorithm represents a solution. In each iteration process, it is necessary to determine the current optimal Y_{best} and worst Y_{worst} solutions of the objective function, based on which the latest position $Y'_{i,j,G}$ of the individual is updated using the following equation:

$$Y'_{i,j,G} = Y_{i,j,G} + rand_1 \times (Y_{best,j,G} - |Y_{i,j,G}|) - rand_2 \times (Y_{worst,j,G} - |Y_{i,j,G}|) \quad (71)$$

where $Y_{i,j,G}$ and $|Y_{i,j,G}|$ represent the j th dimension variable and the absolute value of the i th individual in the G -th iteration, respectively. $rand_1$ and $rand_2$ represent random numbers in the range of $[0, 1]$, $Y_{best,j,G}$ and $Y_{worst,j,G}$ represent the j th dimension variables of the optimal and worst solutions at the G -th iteration.

- (3) Greedy evolution: The individual values, $Y_{i,G}$ and $Y'_{i,G}$, are compared before and after the update, and the solution is selected with a better adaptability to survive to the next generation:

$$Y_{i,G+1} = \begin{cases} Y'_{i,G}, & f(Y'_{i,G}) \leq f(Y_{i,G}) \\ Y_{i,G}, & \text{otherwise} \end{cases} \quad (72)$$

where $Y_{i,G+1}$ and $Y_{i,G}$ represent the i th individual at the $G + 1$ -th iteration and G -th iteration, respectively; $Y'_{i,G}$ represents the updated individual of $Y_{i,G}$, and $f(Y'_{i,G})$ and $f(Y_{i,G})$ represent the updated fitness values, respectively. The aforementioned optimal solution updating method is used to guide other solutions to move to the optimal solution position. However, when solving multi-peak optimization problems, the optimal individual may fall into the local optimal area, causing it to fall into the local optimal solution and converge. The Levy flight has powerful search capabilities, which is helpful in increasing the search range and to avoid falling into local optimal solutions. In this study, the Levy and p-JAYA search methods were combined to enhance the searching and value-finding abilities of the algorithm. A certain individual of the population obtains a new individual via the Levy flight Y_{i+1} .

$$Y_{i+1} = Y_i + \kappa \oplus L(\lambda) \quad (73)$$

Here, κ is the step-size control parameter, $\oplus$ indicates the point-to-point multiplication, and λ is the Levy distribution index, $0 < \lambda \leq 2$, $L(\lambda)$ is the Levy flight function. The updated formula for the step-size is

$$step = \kappa \oplus L(\lambda) \sim 0.1 \frac{u}{|v|^{1/\lambda}} (Y_r - Y_{best}) \quad (74)$$

where Y_{best} is the optimal value of the current iteration, and Y_r is a randomly selected solution in the population that is mutually exclusive with Y_{best} . u and v are random variables with a normal distribution $u \sim N(0, \sigma_u^2)$, $v \sim N(0, \sigma_v^2)$. The values of σ_u and σ_v are as follows:

$$\sigma_u = \left\{ \frac{\Gamma(1+\lambda) \sin(\pi\lambda/2)}{\Gamma[(1+\lambda)/2] \times \lambda \times 2^{(\lambda-1)/2}} \right\}^{1/\lambda}, \sigma_v = 1 \quad (75)$$

where Γ denotes the gamma function. The optimal solution Y_{best} can be obtained based on the aforementioned. The updated formula is as follows:

$$Y_{best}^* = Y_{best} + step \times \delta \quad (76)$$

where Y_{best}^* is the optimal solution after the Levy flight, and $step \times \delta$ is the Levy flight process. $\delta \in rand(0, 1)$ is the step control factor. The following is then calculated:

$$Y_{i,G+1} = \begin{cases} Y_{i,best}, & f(Y_{i,best}) \leq f(Y_{i,best}^*) \\ Y_{i,best}^*, & \text{otherwise} \end{cases} \quad (77)$$

- (4) Output the optimal solution: The iterative update and greedy evolution are repeated until the maximum number of iterations is reached or the convergence condition is met; the identified optimal solution is the output.

Based on the aforementioned derivation, the pseudocode of the improved p-JAYA optimization algorithm is as follows:

4. Simulation results and analysis

4.1. Simulation parameter settings

A simulation was analyzed to verify the performance of the blockchain-based IIoT optimization algorithm. A resource allocation block diagram of the IIoT task data is shown in Fig. 2, demonstrating that when the terminal task arrives, the terminal sends the

Algorithm 1: p-JAYA optimization algorithm

Input: Optimization objective $\Psi(\mathcal{Y})$, $\mathcal{Y} = \{y_1, y_2, \dots, y_j, \dots\}$
 Constraint condition Ω

Output: Optimal fitness value $\psi(\mathcal{Y}_{best})$
 Optimal decision variable $\mathcal{Y}_{best}$

Initialize: Population size $\mathcal{N}$
 Iterations $\mathcal{K}$
 gBest positions $\mathcal{Y}_{best,g}$
 gBest score value $\psi(\mathcal{Y}_{best,g})$
 Best positions $\mathcal{Y}_{best}$
 Best score value $\psi(\mathcal{Y}_{best})$
 Worst positions $\mathcal{Y}_{worst}$
 Worst score value $\psi(\mathcal{Y}_{worst})$
 gFitness matrix $\mathcal{G} = \{\mathcal{G}_1, \mathcal{G}_2, \dots, \mathcal{G}_{\mathcal{K}}\}$
 pFitness matrix $\mathcal{F} = \{\Psi(\mathcal{Y}_1), \Psi(\mathcal{Y}_2), \dots, \Psi(\mathcal{Y}_{\mathcal{N}})\}$
 Positions $\mathcal{B} = \{\mathcal{Y}_1, \mathcal{Y}_2, \dots, \mathcal{Y}_{\mathcal{K}}\}$

```

1  while  $\mathcal{K} \leq \mathcal{K}_{max}$  do
2    while  $\mathcal{N} \leq \mathcal{N}_{max}$  do
3      Pick a random solution  $\mathcal{Y}_{r,\mathcal{K}} \neq \mathcal{Y}_{best,\mathcal{K}}$ 
4      Generate random numbers  $\lambda$  and  $\delta$ , where  $0 < \lambda \leq 2, 0 < \delta < 1$ 
5      Generates the variance of the random variable  $\mu$  and  $\nu$ 
6       $\sigma_\mu^2 = \{\frac{\Gamma(1+\lambda) \times \sin(\pi\lambda/2)}{\Gamma[(1+\lambda)/2] \times \lambda \times 2^{(\lambda-1)/2}}\}^{1/\lambda}$ ,  $\sigma_\nu = 1$ 
7      Generate random variables  $\mu, \nu$  subject to a normal distribution  $\mu \sim \mathcal{N}(0, \sigma_\mu^2), \nu \sim \mathcal{N}(0, \sigma_\nu^2)$ 
8      Calculating step  $step = \kappa \otimes L(\lambda) \sim 0.1 \frac{\mu}{|\nu|^{1/\lambda}}$   $0 < \lambda \leq 2$ 
9      Generate new solution  $\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^* = \mathcal{Y}_{best,\mathcal{K},\mathcal{N}} + step \times \delta$ 
10     if  $\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^* \notin \Omega$ :
11       update  $\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^* \leftarrow \mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*$ 
12     endif
13     Calculate the value of  $\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*, \Psi(\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*)$ 
14     Calculate the value of  $\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*, \Psi(\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*)$ 
15     Update the next generation of solutions  $\mathcal{Y}_{\mathcal{K}+1,\mathcal{N}}$ 
16     if  $\Psi(\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}) < \Psi(\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*)$ :
17        $\mathcal{Y}_{\mathcal{K}+1,\mathcal{N}} \leftarrow \mathcal{Y}_{best,\mathcal{K},\mathcal{N}}$ 
18        $\mathcal{F}_{\mathcal{N}} \leftarrow \Psi(\mathcal{Y}_{best,\mathcal{K},\mathcal{N}})$ 
19     else:
20        $\mathcal{Y}_{\mathcal{K}+1,\mathcal{N}} \leftarrow \mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*$ 
21        $\mathcal{F}_{\mathcal{N}} \leftarrow \Psi(\mathcal{Y}_{best,\mathcal{K},\mathcal{N}}^*)$ 
22     endif
23    $\mathcal{N}_{\mathcal{K}} \leftarrow \text{argmin} \mathcal{F}, \mathcal{B}_{\mathcal{K}} \leftarrow \mathcal{Y}_{\mathcal{N}_{\mathcal{K}}}$ 
24    $\mathcal{F}_{best} \leftarrow \min \mathcal{F}, \mathcal{G}_{\mathcal{K}} \leftarrow \mathcal{F}_{best}$ 
25  $\mathcal{Y}_{best,g} \leftarrow \mathcal{B}_{\text{argmin}(\mathcal{G})}, \psi(\mathcal{Y}_{best,g}) \leftarrow \min(\mathcal{G})$ 

```

remaining resource information and task request information of the terminal to the edge server, which automatically triggers the remaining terminal resource information contract to synchronize with the blockchain platform for other purposes. The server can obtain the relevant information in a timely manner. Simultaneously, the edge server submits the terminal request information to the request contract, verifying the legitimacy of the request information. Subsequently, the request contract obtains the remaining global resource information from the blockchain platform, and submits it, as well as the request information to the optimization algorithm for optimization. The algorithm is executed to obtain the resource allocation results, which are then used to allocate the corresponding resources to minimize the weighted sum of the task-processing delay and energy consumption of the system. After completing the task, each device participating in the calculation submits the transaction information to the blockchain platform. The transaction includes detailed information of the transaction and the reward value obtained to complete the calculation.

The following main parameters were set during the simulation: there were 100–150 terminals, the arrival of the terminal tasks obeyed the Poisson distribution, there were 30–60 edge servers, the size of the tasks generated by the terminal ranged between 1.5 and 2.5 M, the computing power of the terminal ranged between 10–30 GHz, and that of the edge-computing server ranged between 100–300 GHz. In this paper, we set the simulation parameters based on several related studies such as the research results on the

Table 3
Simulation parameters.

Symbol	Parameter	Value
D_{num}	Number of terminals	[100,150]
E_{num}	Number of edge computing servers	[30,60]
STR	Task data size	[1.5,2.5] M
TCF	Terminal computing ability	[10, 30] GHz
MSF	Edge server computing ability	[100, 300] GHz
BTS	bandwidth	[2.5, 5.5] M
PW	Terminal transmit power	[25, 75] W
TG	channel gain	$[10^{-13}, 10^{-11}]$
β	Energy consumption coefficient	[1, 10]
ξ	weight coefficient	[0, 1]

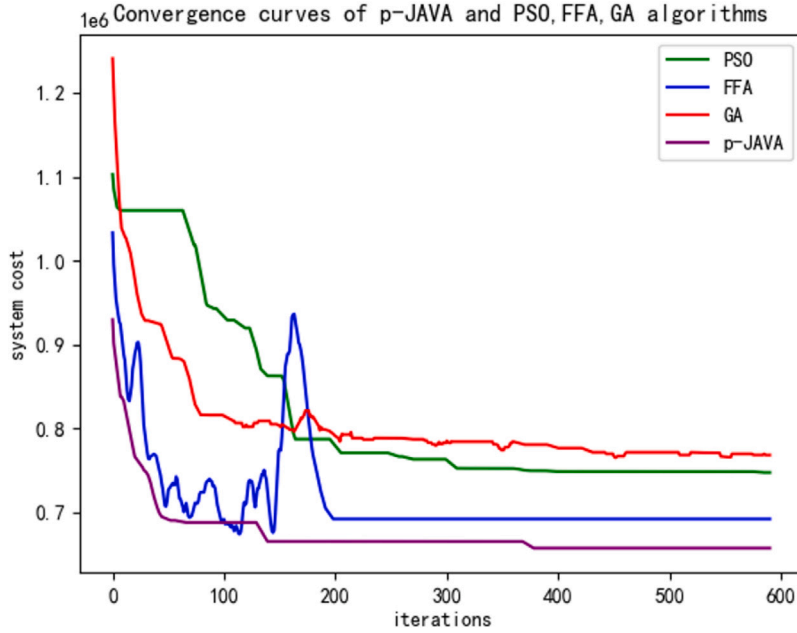


Fig. 3. The convergence process of the improved p-JAYA optimization algorithm and the other three optimization algorithms.

combination of blockchain and industrial IoT in the literature [27] and the research results on the resource allocation of industrial IoT based on blockchain in Ref. [28]. The detailed simulation parameters are listed in Table 3.

4.2. Algorithm performance analysis

To achieve a performance comparison, we simulated the PSO, FFA, and GA algorithms using the same set of parameters.

Fig. 3 presents the convergence process of the improved p-JAYA hybrid optimization algorithm, PSO, FFA, and GA for solving the problem. The abscissa represents the number of iterations and the ordinate represents system consumption. The four algorithms apparently converged at approximately 200 iterations. However, p-JAYA with the improved optimization capabilities was the strongest, and its curve was smoother and not significantly volatile. This indicates that the p-JAYA optimization algorithm is relatively stable while finding solutions during optimization, which reflects its advantages.

Fig. 4 presents a comparison diagram of the load-balancing capabilities of the improved p-JAYA optimization algorithm, PSO, FFA, and GA. The abscissa represents the entry. There are four types of execution devices: the main server, adjacent server, IoT device, and cloud server. The ordinate represents the proportions of each execution method. Each optimization algorithm is represented by a specific color. Overall, the execution methods of the adjacent servers and devices accounted for a relatively large proportion of all algorithms, among which the device execution accounted for the largest proportion. This is mainly because when the device is executing, data can be directly transmitted between devices without being forwarded through the main server, which reduces the transmission delay and system efficiency. Therefore, all the algorithms have a relative preference for device execution. The proximity server cluster execution method was in second place. When the proximity server method calculates the data, it must be forwarded through the main server, and the delay is relatively larger. The system consumption was also relatively high. The proportion of adjacent servers was higher than that of the main server because the main server has fewer resources, and

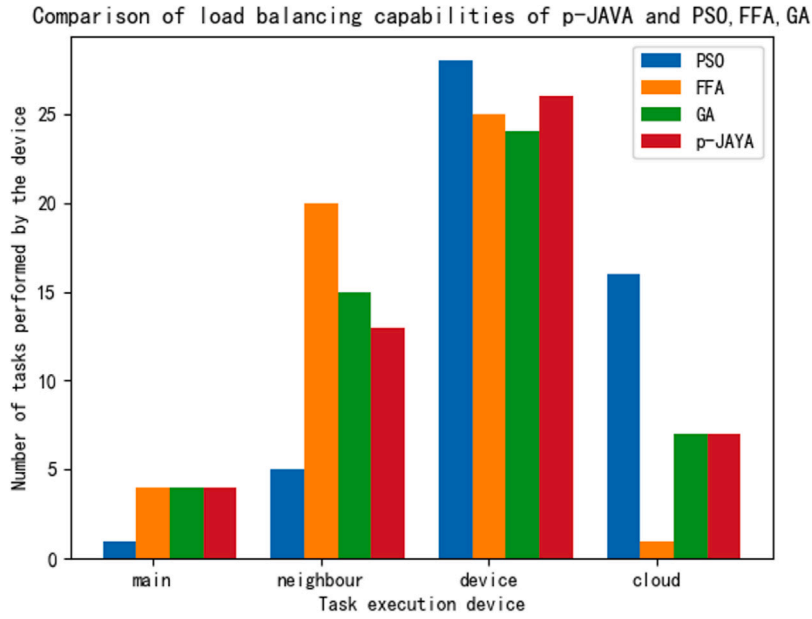


Fig. 4. Comparison of the load balancing capabilities between the improved p-JAYA optimization algorithm and comparison algorithm.

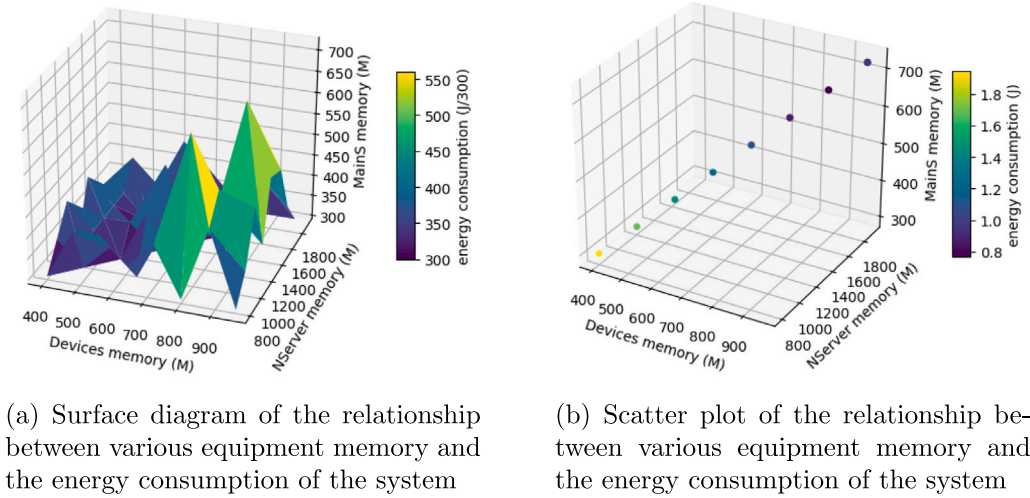


Fig. 5. The relationship between various equipment resources and the energy consumption of the system.

the adjacent server cluster has relatively large resources. The number of resources of the main server limits the amount of data offloading; therefore, the algorithm chooses to close the server for offloading. Based on the offloading proportion of each algorithm, the improved p-JAYA optimization algorithm exhibits better load-balancing capabilities, and balances each execution method such that it can obtain a suitable offload quantity.

Fig. 5 presents the relationship between the number of resources the various execution methods have and the energy consumption of the system when optimizing the problem using the improved p-JAYA optimization algorithm. The abscissa represents the number of resources in the adjacent server and the ordinate represents the terminal device. The vertical coordinates represent the resource size of the main server. Fig. 5(a) presents a surface diagram of the relationship between the amount of various equipment resources and the system consumption. The depth of the color represents the amount of the system consumption. A darker color indicates a lower system consumption. As shown in Fig. 5(a), as the resources of the various devices increased, the weighted sum of the system energy consumption and time consumption gradually decreased. This mainly occurred because when there is a limited number of equipment resources, the constraints are more stringent, and the maximum value of the objective function is significantly constrained. For example, if the amount of equipment resources is excessively low, then the optimization algorithm must find other execution methods and transfer from the device execution to other execution methods, increasing latency and the energy

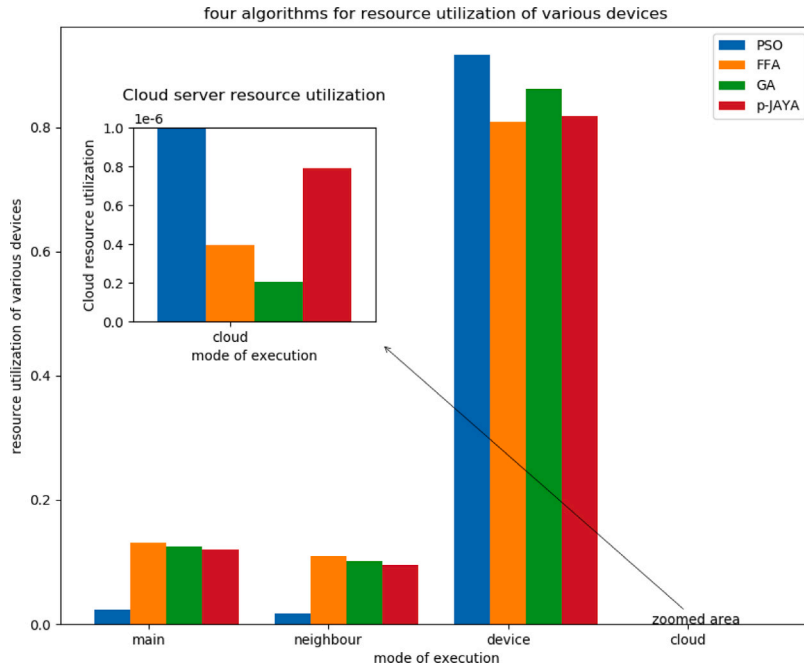


Fig. 6. Improved p-JAYA optimization algorithm and the resource utilization of each device of the three algorithms.

consumption, and resulting in an increased system consumption. Fig. 5(b) presents a three-dimensional scatter diagram of the relationship between the terminal equipment, adjacent servers, main server resources, and the system energy consumption. The figure also demonstrates that the system energy consumption decreases as the amount of various equipment resources increases. When the ratio of the amount of data to the total amount of various equipment resources reaches a certain level, it is difficult for the optimization algorithm to find the global optimal solution of the function. At this time, the algorithm exhibits volatile characteristics.

Fig. 6 presents the resource utilization of various devices when allocating resources between the improved p-JAYA optimization algorithm and the other three algorithms. The abscissa represents various devices, the ordinate represents the resource utilization of various devices corresponding to the algorithm. As shown in the figure, the resource utilization of the terminal devices is significantly higher than that of other devices. This is because when terminal clusters under the same main server transmit data to one another, the transmission delay is significantly smaller than the delay in data transmission to adjacent servers, cloud servers, and other devices. This causes the algorithm to preferentially select the end device as an offload point. However, owing to the small amount of terminal equipment resources and the load-balancing characteristics of the algorithm, part of the data is offloaded to the main server, adjacent servers, and cloud servers. Therefore, the results demonstrate that the resource utilization of the terminal is significantly higher than that of other devices.

4.3. Security analysis

This study introduced a blockchain system for the resource allocation of IIoT. When allocating resources, the equipment participating in resource allocation will have an identification $\{PK_{d_i}, SK_{d_i}, Cert_{d_i}\}$. During the resource allocation process, the terminal device will first use its own private key SK_d to sign the data, which ensures the authenticity and reliability of the data source and prevents harmful data from damaging the system. Second, after the calculation is complete, According to formula (47), the device responsible for the calculation will generate a transaction of the calculation.

The transaction includes information regarding the data, calculation of the private key of both parties, public key, signature, remuneration to be paid, and timestamps. Both parties involved in the calculation sign the transaction, which is then stored in a blockchain system, and ensures the traceability, trustworthiness, and authenticity of the calculation. Simultaneously, the remaining resource information of various devices is synchronized with the blockchain system. By exploiting the shared nature of the blockchain system, multiple node data are consistent, ensuring the authenticity of the system resource information obtained during resource allocation. This helps the optimization algorithm make more accurate decisions and improves the system efficiency. Second, the blockchain system will store the request information and remaining resource information of the terminal. By checking the request information of terminals, we can know the regions with high data traffic and high resource demand, and by checking the remaining resource information, we can know the regions with free resources, and then according to the information of resource demand and free resources, we can allocate resources to the data processing requests of terminals, which can improve the resource utilization rate of various computing devices, balance the pressure between computing devices, and ease the pressure of computing devices in the regions with high data volume. This can improve the resource utilization rate of various computing devices, balance the pressure between computing devices, and relieve the pressure of computing devices in areas with high data volume.

5. Conclusions

Considering the characteristics of the terminal equipment in IIoT, such as a low computing power and storage capacity, a resource allocation framework based on a cloud–edge–device integrated computing framework was proposed in this study. First, the traditional p-JAYA optimization algorithm was improved. The experiments demonstrated that compared with classic optimization algorithms, such as PSO and GA, the improved p-JAYA optimization algorithm demonstrated significantly improved optimization capabilities and notably reduced the energy consumption of the system. Second, considering the problems of data security and the imperfect incentive mechanism in the resource allocation of IIoT, the blockchain system for the resource allocation of IIoT was introduced in this study. For each calculation of the resource allocation, after the calculation is complete, a transaction consisting of the relevant messages for this calculation is submitted to the blockchain system, which saves the transaction information, ensuring the traceability of the calculation process, calculation information sharing, and security of the data. Furthermore, blockchain transactions also include rewards that must be paid to computing participants. This computing framework provides a safe, trustworthy, traceable, and incentive-based model for IIoT resource allocation. Combined with the improved performance-efficient p-JAYA optimization algorithm, IIoT resource allocation provides a relatively good solution. Blockchain is a decentralized, trusted distributed database system, and the combination of blockchain and IoT resource allocation can provide a secure, reliable, and traceable environment for IoT data, which has a wide range of application prospects. However, the present conclusions are drawn from simulations under specific assumptions, and the main issues are also considered when performing modeling. The heuristic p-JAYA algorithm is used in performing the solution of the optimization problem. Heuristic p-JAYA algorithm may have small volatility in finding the solution. In this paper, a trade-off is made between the stability of the algorithm and its exploratory nature in solving the optimization problem using p-JAYA. Nevertheless, this conclusion may have some limitations in practical applications.

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CRediT authorship contribution statement

Jie Lu: Visualization, Validation, Software, Methodology, Conceptualization. **Yuexia Zhang:** Methodology, Conceptualization. **Taifu Yuan:** Software.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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